

Supporting Information for: Early leafout extends calendar but not thermal growing seasons

June 16, 2026

1 Tables

Species	functional group	<i>n</i>
* <i>Acer pensylvanicum</i>	tree	NA
* <i>Acer spicatum</i>	tree	NA
<i>Alnus incana</i>	shrub	31
<i>Amelanchier canadensis</i>	shrub	6
<i>Aronia melanocarpa</i>	shrub	12
<i>Betula alleghaniensis</i>	tree	24
<i>Betula papyrifera</i>	tree	13
<i>Betula populifolia</i>	tree	24
<i>Diervilla lonicera</i>	shrub	16
<i>Myrica gale</i>	shrub	15
* <i>Quercus alba</i>	tree	NA
* <i>Quercus rubra</i>	tree	NA
<i>Sambucus racemosa</i>	shrub	11
<i>Sorbus americana</i>	shrub	5
<i>Spiraea alba</i>	shrub	19
<i>Spiraea tomentosa</i>	shrub	21
* <i>Vaccinium myrtilloides</i>	shrub	NA
<i>Viburnum cassinoides</i>	shrub	25

Table S1: Common garden focal species. In 2017, plantings were randomized between 16 plot blocks. Individuals that were too small to survive outside were maintained in the growth facilities for an additional year and out-planted in the early spring of 2018. Plots were divided between tree plots which included species *Acer pensylvanicum*, *Amelanchier canadensis*, *Alnus incana*, *Betula papyrifera*, *Betula populifolia*, *Beluta alleghaniensis*, *Quercus alba*, and *Quercus rubra* and shrub plots which included the remaining species and shade cloth. * indicates species that were planted but not included in analyses due to low survivorship.

2 Figures

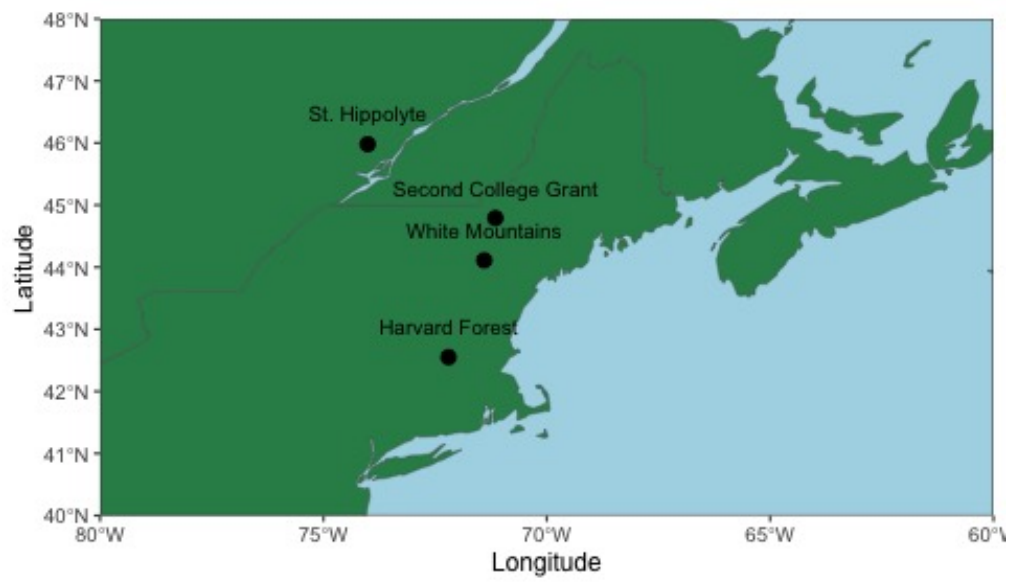


Figure S1: A map of the four site sampled in this study

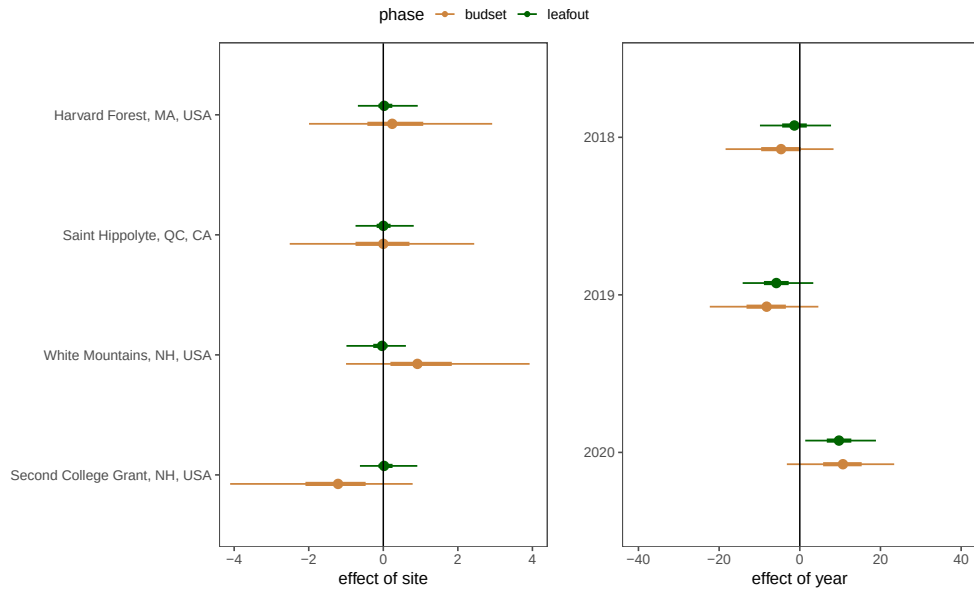


Figure S2: Difference in leafout and budset partitioned between populations (a) and years (b). Points represent the median effect size estimate, and thick and thin bars the 50% and 90% uncertainty intervals.

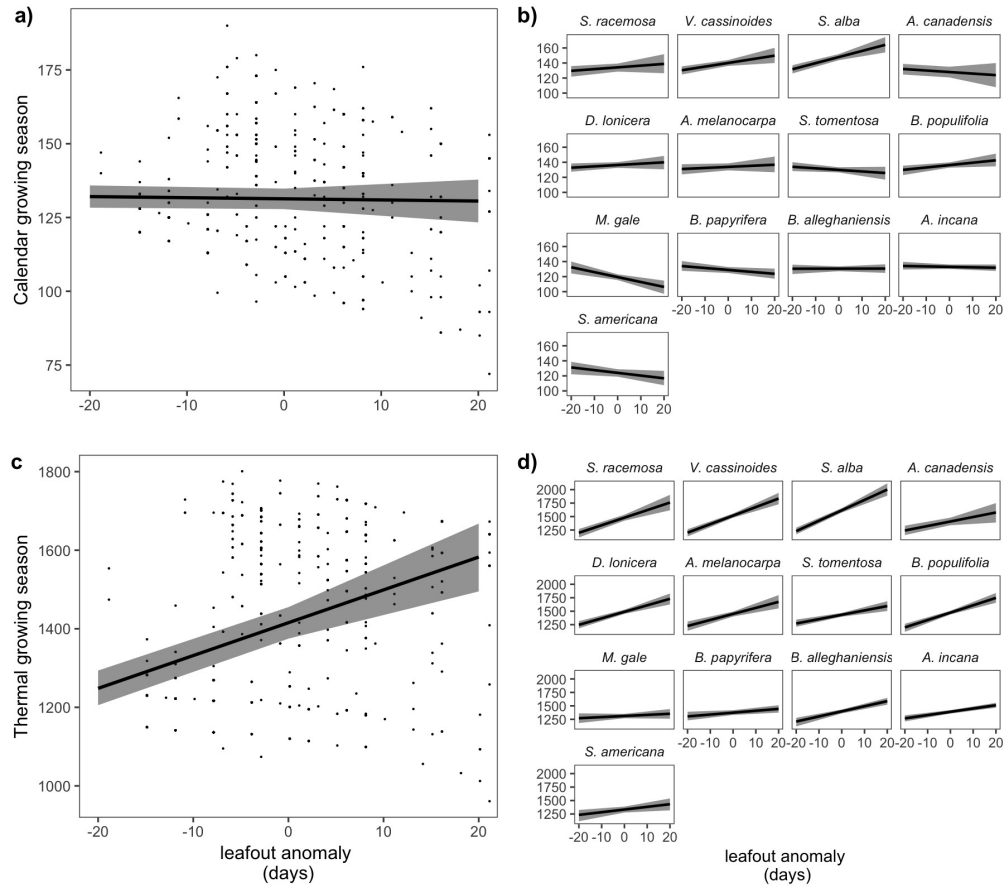


Figure S3: The relationship between the day of leafout and growing season length (defined by leaf coloration) differs between the calendar growing season and the thermal growing season. Later leafout did not affect calendar growing season (a) but this pattern varied across species in our study (b). Increasingly later leafout resulted in a longer thermal growing season (c) though this effect was stronger for species that typically leafout earlier in the season—panels in c) display in the typical order of leafout among species. The trend lines represent the median effect of leafout timing on growing season length and shaded bars the 90% uncertainty interval. Points in a), and c) represent the raw data.

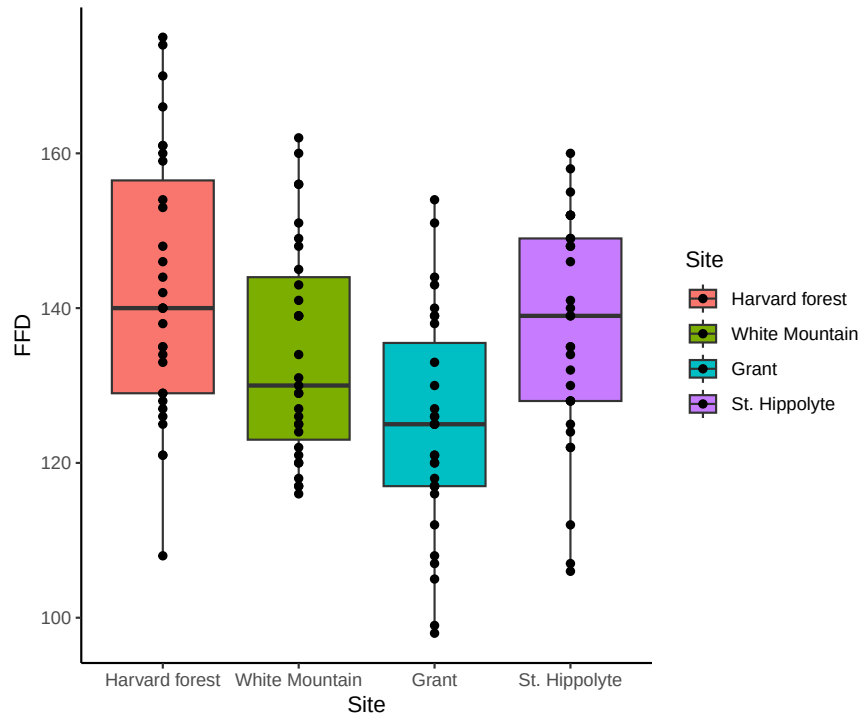


Figure S4: Differences in frost free days (FFD) between sites. Each point represents the number of days between the 90th percentile of last spring frost days and the 90th percentile of first fall frost days for a given year between 1980 and 2010. To calculate FFD, we used the long-term gridded estimates of minimum temperature from Daymet (version 4, 1).

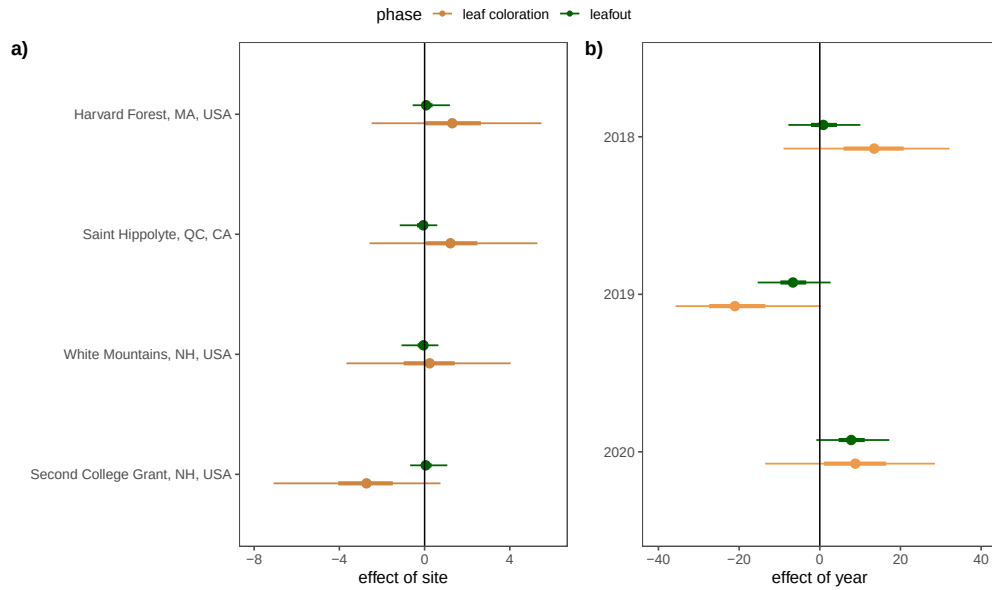


Figure S5: Difference in leafout and leaf coloration partitioned between populations (a) and years (b). Points represent the median effect size estimate, and thick and thin bars the 50% and 90% uncertainty intervals.

References

- [1] Thornton, M., Shrestha, R., Wei, Y., Thornton, P., and Kao, S.-C. Daymet: Daily surface weather data on a 1-km grid for north america, version 4, (2020).